

Aditya Kulkarni

adityakulkarnius@gmail.com | (281) 876-7066 | linkedin.com/in/aditya-kulkarni-355b81217 | kulkarniaditya.com | Raleigh, NC

PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with production experience building full-stack microservices, RESTful APIs, and AI-integrated backend systems. Shipped 10+ features across 7 production deployments at an AI-native startup using TypeScript, Node.js, PostgreSQL, and OAuth 2.0 (Auth0). Built Spring Boot 3.5 enterprise APIs in Java 21 with PostgreSQL row-level locking, Flyway migrations, and Testcontainers integration tests. Proficient in React and Next.js frontends, GitHub Actions CI/CD, Docker, and multi-model AI agent tooling across Claude, GPT-4o, and Gemini.

CORE COMPETENCIES

Spring Boot Microservices (Java/Kotlin)

TypeScript + React / Next.js

RESTful API Design

PostgreSQL + Flyway Migrations

OAuth 2.0 / Auth0

AI Agent Tooling (Claude, GPT-4o, Gemini)

Docker + Kubernetes + GitHub Actions CI/CD

Testcontainers + Integration Testing

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern -- Raleigh, NC

- Developed **10+ backend features and RESTful APIs** for an AI-powered claims automation platform using TypeScript, Node.js, Express, PostgreSQL, and Prisma -- 7 production deployments with zero regressions
- Implemented **OAuth 2.0 authentication via Auth0**, role-based access control, real-time event pipelines with Redis and Pusher, and document processing pipelines (PDF, email parsing via Microsoft Graph API, S3)
- Integrated **multi-model AI agent tooling** across 3 providers and 8+ models (Anthropic Claude, OpenAI GPT-4o, Google Gemini) using MCP; built Temporal.io workflow orchestration for multi-step agentic pipelines
- Shipped Sentry observability, Playwright automated testing, and Docker containerization; contributed **394 commits** and participated in full PR review cycle on a 2-person engineering team

NC State University April 2025 - Present

Graduate Research Assistant -- Raleigh, NC

- Designed a **cloud-based data platform** with JBrowse 2, serving 500GB genomic datasets to 100+ researchers worldwide on Google Cloud Run
- Collaborated with ML research team on model architecture design, hyperparameter tuning, and time-series analysis -- improved model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern -- Remote

- Built 5+ AI-driven automation workflows including a health claim tool using OCR, NLP, and ML pipelines; improved claim accuracy by 25%
- Automated deployments on AWS and GCP; integrated monitoring dashboards with Grafana; supported model validation and red-teaming

Capgemini Jun 2023 - Aug 2023

AI/ML Intern -- Remote

- Led end-to-end development of a distributed data pipeline using Hadoop + Spark and Kafka streaming, reducing ticket-processing latency by 20%

PROJECTS

Inventory Reservation Service Spring Boot + PostgreSQL

Built a transactional inventory reservation API in **Java 21 + Spring Boot 3.5** preventing overselling via PostgreSQL row-level locking (SELECT FOR UPDATE) across multi-SKU reservations. Idempotent POST endpoints with RFC 9457 error responses, automated expiry via scheduled cleanup job. Verified concurrency correctness with **Testcontainers integration tests** -- two concurrent reservations against quantity 1 produce exactly one success and one 409 Conflict. Flyway migrations, Spring Security (role-based), multi-stage Docker build, GitHub Actions CI.

Stack: Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Spring Data JPA, Spring Security, Testcontainers, Docker | github.com/Aditya-k24/InventoryReservationService

PulseRank -- Real-Time Leaderboard & Activity Feed Full-Stack + Analytics

Built an event-driven leaderboard platform with **NestJS + Next.js + TypeScript**: MySQL for transactional writes with idempotency-key deduplication and outbox pattern, MongoDB for feed projections, Redis sorted sets for sub-millisecond leaderboard reads, live score updates over Server-Sent Events. Validated multi-store consistency with Testcontainers; benchmarked hot paths with k6; shipped observability with OpenTelemetry + Prometheus and CI/CD with GitHub Actions.

Stack: TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry | github.com/Aditya-k24/PulseRank

LLM Eval Harness -- Vendor-Neutral Benchmark AI Agent Tooling

Built a benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across 2,000 frozen test examples. Tracked ExactMatch, TokenF1, abstention accuracy, grounded hallucination rate, p50/p95 latency, and cost/call. Applied statistical significance via paired bootstrap CIs, McNemar test, and Holm correction -- demonstrating practical **AI coding agent and evaluation pipeline** expertise.

Stack: Python, Anthropic/OpenAI/Google GenAI SDKs, DuckDB, Parquet, Streamlit, Docker | github.com/Aditya-k24/llm-eval-harness

Glance AI -- Privacy-First RAG System AI Backend

Built a compliance-ready RAG system for fintech/healthcare with per-tenant data isolation at the vector retrieval layer, preventing context leakage across clients. Designed deterministic output architecture for regulated data environments.

Stack: Python, LLMs, Vector Databases, RAG | github.com/Aditya-k24/GlanceAI

EDUCATION

MS Computer Science -- North Carolina State University Aug 2024 - May 2026

Raleigh, NC | Courses: Artificial Intelligence, Operating Systems, Data Science, Algorithms, Software Engineering

B.Tech Computer Science -- Mumbai University Aug 2020 - May 2024

Mumbai, India | CGPA: 9.32/10 | Courses: Big Data, Machine Learning, DS & Algorithms, Software Engineering

SKILLS

Backend: Java, Kotlin, Spring Boot, Node.js, Python, TypeScript, PostgreSQL, MySQL, MongoDB, Redis, Prisma

Frontend: React, Next.js, NestJS, Tailwind, Angular

AI/ML: LLM APIs (Anthropic, OpenAI, Gemini), MCP, LangChain, RAG, TensorFlow, PyTorch, Hugging Face

DevOps: Docker, Kubernetes, GitHub Actions, AWS, GCP, Grafana, OpenTelemetry, Prometheus, Sentry **Testing:** Testcontainers, Playwright, Jest, k6